# Ethan Wu

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**Looking for summer 2027 intern opportunities for compiler and systems level engineering**

## EDUCATION

**Northwestern University** **September 2026 – June 2030 (Anticipated)**
*Computer Science & Operations Research (IEMS)* *Evanston, IL*

- **NU Formula Racing (FSAE):** Firmware engineer; using C to build high voltage battery management system.

## SKILLS

**Languages:** Python, C++, Mojo, Rust, Go, JavaScript
**Tooling & Frameworks:** PyTorch, LLVM, MLIR, Cranelift, Google OR-Tools, Jax/Flax, LightGBM, Optuna, Docker
**Knowledge/Domains:** Multi-Threading, Optimizations, Clustering Algorithms, ML, Git, Linux, SIMD, C-FFI

## WORK EXPERIENCE

**Research Intern** **October 2024 – Present**
*MIT Jameel Clinic* *Cambridge, MA*

- Performed clustering analysis on public health data; **co-authoring 2 in-progress papers** targeting PLOS One.
- Developed novel method to find the best clustering from clusterings generated with different hyperparameters.
- Replicated & verified cancer prediction computational models in published papers.
- Co-designed **a custom featurizer** for a cancer prognostic machine learning model with an MIT PostDoc.
- Had several of my projects featured in an MIT blog post: *See Blog Post*

## PROJECTS & OPEN SOURCE CONTRIBUTIONS

**Mojo Compiler & STL [Contributor] - MLIR, Mojo, C++, LLVM** *https://github.com/modular/modular/commits?author=ethqnol*

- Resolved false-positive infinite recursion warnings in Mojo's semantic CFG pass (LowerSemanticCF) by tracking HLCF loop break operations across nested subregions.
- Fixed LLVM codegen bug by mapping SIMD comparisons to FCMP UNE, correcting vector NaN evaluation.
- Wrote MLIR regression and Mojo integration test suites to validate control flow lowering and IR emission.

**polars-mojo [Author] - Mojo, Apache Arrow, SIMD, Polars** *https://prefix.dev/channels/modular-community/packages/polars-mojo*

- Built zero-copy interop between Mojo and the Rust Polars dataframe library via Apache Arrow C Data Interface
- Implemented an RAII wrapper to synchronize Mojo `__deinit__` with Arrow C release callbacks to prevent leaks.
- Implemented SIMD reduction kernels (sum, mean, std) via Mojo's vectorize primitives.

**Strata [Author] - Mojo, LAPACK/BLAS, SIMD** *https://prefix.dev/channels/modular-community/packages/strata*

- Developing a pure Mojo machine learning library with Sklearn ergonomics and zero Python runtime dependencies.
- Leveraged strict traits, static polymorphism, and generic SIMD types across float/int bit-widths to optimize ML models (Linear models, Random Forests, K-Means). Average of 1.9 times faster than Sklearn. *See benchmarks*
- Applied LAPACK/BLAS to handle linear algebra computation.

**Salmon Programming Language [Author] - C++ 20, LLVM, Clang** *https://github.com/ethqnol/salmon*

- Built an end-to-end compiled language called Salmon, that targets native machine code via LLVM IR
- Implemented SSA code-gen using LLVM C++ API with scope management & dynamic list runtime structs.
- Designed deterministic resource cleanup via a LIFO `defer` unwinding stack.

**Kluster-Fudge (PyPI) [Author] - Python, JIT, GPU** *https://pypi.org/project/kluster-fudge/*

- A Python library for clustering categorical data using K-Modes with various distance metrics (**3000+ downloads**).
- Implemented multi-threading, GPU acceleration, and JIT optimizations for better performance.

**CubeCL (TracelAI) [Contributor]** *https://github.com/tracel-ai/cubecl*

- Contributed upstream bug fix to CubeCL, a multi-platform GPU compute language extension for Rust.

**Modular Community & Prefix Ecosystem (Modular AI) [Contributor]** *https://github.com/modular/modular-community/pull/348*

- Resolved critical packaging automation failure by debugging and patching CI/CD scripts for the Mojo toolchain.

